

Practical:4

Singly Linked List: Dynamic Insertion, Deletion, and Traversal

Practical Aim

To implement core singly linked list operations and develop the ability to manipulate dynamic sequences through real-world problem scenarios.

Problem 1

A hospital manages a queue of patient tokens. New critical patients must be added to the front, routine patients are added to the end, and occasionally a patient with a priority number must be inserted at a specific position in the queue. Given a sequence of such operations, implement all three insertion types and print the final queue after each operation. Describe how your insertion at a specific position works. What does your solution do if the position given is greater than the current length of the queue? Is that handled explicitly or does it cause an error?

CODE:

```
package prac4;

import java.util.LinkedList;

public class HospitalQueueProblem1 {

    public static void main(String[] args) {

        LinkedList<Integer> queue = new LinkedList<>();

        queue.addLast(101);
        System.out.println("Queue: " + queue);

        queue.addLast(102);
        System.out.println("Queue: " + queue);

        queue.addFirst(100);
        System.out.println("Queue: " + queue);

        int position = 2;
        int patientToken = 105;
```

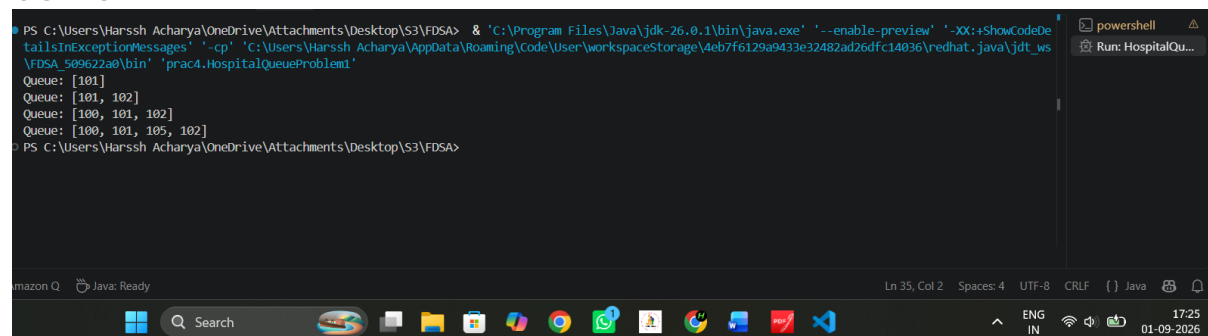
```

        if (position <= queue.size()) {
            queue.add(position, patientToken);
        } else {
            queue.addLast(patientToken);
        }

        System.out.println("Queue: " + queue);
    }
}

```

OUTPUT:



```

PS C:\Users\Harssh Acharya\OneDrive\Attachments\Desktop\S3\FDSA> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDe
tailsInExceptionMessages' '-cp' 'C:\Users\Harssh Acharya\AppData\Roaming\Code\User\workspaceStorage\4eb7f6129a9433e32482ad26dfc14036\redhat.java\jdt_ws
\FDSA_509622a0\bin' 'prac4.HospitalQueueProblem1'
Queue: [101]
Queue: [101, 102]
Queue: [100, 101, 102]
Queue: [100, 101, 105, 102]
PS C:\Users\Harssh Acharya\OneDrive\Attachments\Desktop\S3\FDSA>

```

Problem 2

The hospital also needs to remove a patient token from the queue when a patient leaves, print all remaining tokens from last to first for an end-of-day audit, and display the full queue from front to back at any point. Implement deletion by value, reverse printing, and forward traversal on the same queue from Problem 4.1a. Describe how your reverse printing works. Does it use any extra storage, or does it work purely by following the links? Can you think of another way to reverse-print without modifying the original queue?

CODE:

```

package prac4;

import java.util.LinkedList;

public class HospitalQueueProblem2 {

    public static void main(String[] args) {

        LinkedList<Integer> queue = new LinkedList<>();

        queue.add(101);
        queue.add(102);
        queue.add(103);
        queue.add(104);
    }
}

```

```
System.out.println("Queue from Front to Back:");
for (int token : queue) {
    System.out.print(token + " ");
}

int patientToRemove = 102;
queue.remove(Integer.valueOf(patientToRemove));

System.out.println("\n\nAfter removing " + patientToRemove + ":");
System.out.println(queue);

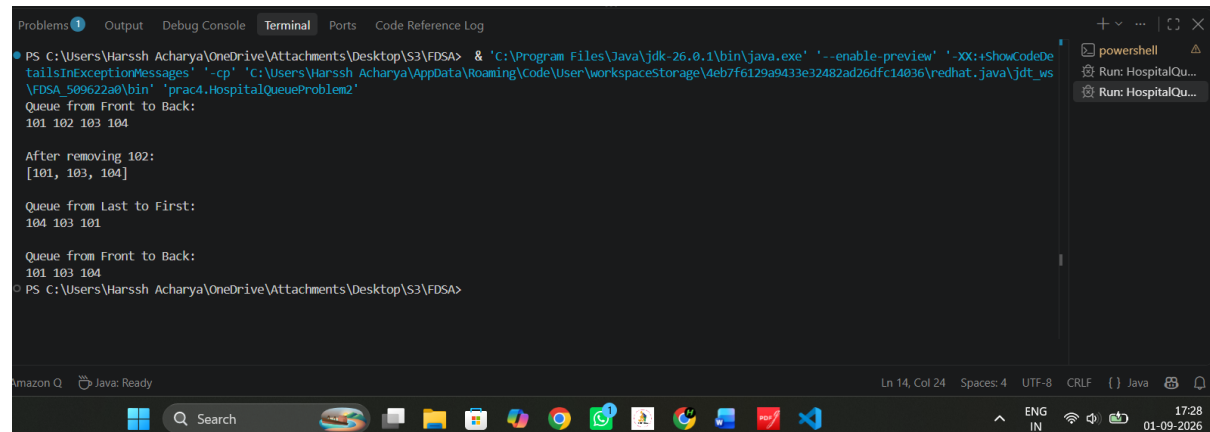
System.out.println("\n\nQueue from Last to First:");

for (int i = queue.size() - 1; i >= 0; i--) {
    System.out.print(queue.get(i) + " ");
}

System.out.println("\n\nQueue from Front to Back:");

for (int token : queue) {
    System.out.print(token + " ");
}
}
```

OUTPUT:



```
PS C:\Users\Harssh Acharya\OneDrive\Attachments\Desktop\S3\FDSA> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-enable-preview' '-xx:+ShowCodeDe
tailsInExceptionMessages' '-cp' 'C:\Users\Harssh Acharya\AppData\Roaming\Code\User\workspaceStorage\4eb7f6129a9433e32482ad26dfc14036\redhat.java\jdt_ws
\FDSA_509622a0\bin' 'prac4.HospitalQueueProblem2'
Queue from Front to Back:
101 102 103 104

After removing 102:
[101, 103, 104]

Queue from Last to First:
104 103 101

Queue from Front to Back:
101 103 104
PS C:\Users\Harssh Acharya\OneDrive\Attachments\Desktop\S3\FDSA>
```

GitHub link: <https://github.com/haarshaacharya/FDSA/tree/main/prac4>